



Modding

This page discusses the basics of modding. For information about using game mods, see [Mod](#)

This article is [timeless](#) and should be accurate for any version of the game.

Modding (using 'mods', short for modification) in Europa Universalis V provides players with the ability to alter various aspects of the game or incorporate new features, enhancing the overall gaming experience. While certain elements remain hardcoded and cannot be modified, such as the Migration or Battle casualty logic or map modes, a wide range of options are available for customization.

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Getting started

Setting up the mod folder

See also: [Mod structure](#)

- Create a new folder in Documents/Paradox Interactive/Europa Universalis V/mod, this new folder corresponds with the game folder in the base game.
 - Don't directly modify the game files, they will be overwritten with any game update.
- In the new folder, create your "in_game", "main_menu", and "loading_screen" top level folders, then within those "common", "localization", etc. folders of your mod as needed.
 - Follow the same structure as found in Europa Universalis V\game folder.
- Place .txt and other files for the actual content inside the appropriate folder.
- Add a .metadata folder with a metadata.json file inside. The file may be bare bones, but is a requirement in order for the game to load your mod at all.

The in-game mod tools can also be used to set up the basic mod folder. The images for what that looks like.

Basic tools and resources

- Utilize a source code editor with syntax highlighting and other helpful features, such as Code Search. Popular free programs for modding Europa Universalis V include:
 - Visual Studio Code (<https://code.visualstudio.com/>) has the extension Paradox Highlight and CwTools (<https://marketplace.visualstudio.com/items?itemName=toby.cwtools-vscode>) available.
 - IntelliJ (<https://www.jetbrains.com/idea/download/other.html>) (The Community Edition is free) has the extension Paradox Language Support (<https://plugins.jetbrains.com/plugin/16825-paradox-language-support>) available.



The mod tools button is immediately next to the mod manager button



The create mod screen prompts most basic information a mod should have. It creates an empty folder in the local mods document folder.

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- There is a community-sourced custom language for Europa Universalis V that can be found on github (https://github.com/Aetherum17/EU5_Notepad_Language).
 - Alternatively, select *Perl* as your language syntax for closet built-in syntax highlighting.
- Add the following options in Steam via left click on Europa Universalis V -> Properties at the bottom of the General folder:
 - **-debug_mode** to activate the console and in-game dev tools. Alternatively, there are also mods to activate/deactivate debug mode in-game. Also, hotloads changes to your mod files without needing to restart the game.
 - A few files are not covered, such as `on_action`.
- Use **script_docs** and **dump_data_types** commands in the game console to generate other logs with effects, triggers, and scopes.
 - Access the console in-game with ~ when debug_mode is active.
 - Generated files from **script_docs** are located in Documents\Paradox Interactive\Europa Universalis V\docs; see also [Effect](#), [Trigger](#), [Event target](#), and [Modifier types](#) on the wiki
 - Generated files from **dump_data_types** are located in Documents\Paradox Interactive\Europa Universalis V\logs\data_types; see also [GUI script](#) on the wiki
 - This is the most reliable technical documentation you can get.
 - Those extracted files can also be found on GitHub (<https://github.com/Europa-Universalis-5-Modding-Co-op/modding-digests>)
- The [Modding Git Guide](https://docs.google.com/document/d/1bQdOVMY6FTu-2AKXZbIYp6bF2-_W2JMUtXc5a0nZ8Ls) (https://docs.google.com/document/d/1bQdOVMY6FTu-2AKXZbIYp6bF2-_W2JMUtXc5a0nZ8Ls) is a community made guide for using Git, GitHub/GitLab, and related tools such as KDiff3. It can be a useful stop for questions beyond this wiki, and contains step-by-step guides for much of what is talked about here. Though the examples are HOI4 based, the principles apply equally well to any Paradox game mod.

Best practices

- Use proper indentation to ensure opening and closing brackets are on the same level. This allows for code folding, easier error spotting, and improved readability.
 - Consider using an automatic formatting extension for your editor.
- When uncertain about implementation, search the base game code for relevant keywords using your editor.
- Add comments, starting with #, to remember later how things works, to navigate easier or to disable parts of code without deleting them.
- Avoid overwriting items you didn't change to minimize conflicts and enhance clarity.

Priority of modded files

Main article: [Mod files load order](#)

- Using a filename that exists in vanilla will overwrite the original file completely.
- If several files with the same name are used by different mods, the lowest mod list position in the launcher is relevant.



- This takes priority over the position in the mod list in the launcher.

Debugging

- Read the error.log to find exceptions and warnings the game identified in Documents/Paradox Interactive/Europa Universalis V/logs
- **-debug_mode** adds a button that opens the `error.log` for you and also shows its current number of lines.
 - The default editor used for this on Windows is Notepad, but it is configurable.
 - You can either edit Documents/Paradox Interactive/Europa Universalis V/pdx_settings.json directly, adding the following keys under the System key:
 - "editor": "C:\\Program Files\\Sublime Text\\sublime_text.exe" - tells the game which editor to use, you can use any valid editor on your computer, just make sure to use the correct path.
 - "editor_postfix": " :\$:1" - this is used when opening the editor, for example this Sublime Text specific postfix makes sure the editor is opened scrolled to the latest line.
 - Or you can use the ingame settings editor accessible through the in-game console, under the Settings button. Choose System in the Category selector and find the Editor key.
- **--ignore-disable-mods-on-crash** . If the game crashes during loading resources, it will load with all mods disabled the next time. Use this option to force load the same playset. Useful when debugging crash in your own mod.
- Backup your work, things can go wrong, and it is always good to have a backup.
 - Using a version control system like **Git** (<https://git-scm.com/>) allows you to track your changes, revert to a working version and collaborate.
 - If you are part of a team, consider using GitHub for team collaboration. If you are new to Git, a program like [Github Desktop](https://desktop.github.com/) (<https://desktop.github.com/>) is great for beginners.
- Set *encoding* to **UTF-8 with BOM** (<https://en.wikipedia.org/wiki/UTF-8>) inside your editor for code files, especially for localization .yaml files.
- For localization files, make sure your filenames end with **L_language** like **foo_1_english.yaml**. Note the first character in the suffix is an ell (lowercase L), not the numeral 1 or uppercase i.
- Join the **EU5 Mod Coop discord** (<https://discord.gg/KeXMv4yM3a>) to ask other modders for help.

Tools and utilities

Add your mod tool to the wiki



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Paradox Highlight	Editor (VS Code)	https://marketplace.visualstudio.com/items?itemName=dragon-archer.paradox-highlight	Up to date syntax highlighting for all Europa Universalis V files. Other paradox games like Crusader Kings III are also supported.
CWTools - Paradox Language Services	Editor (VS Code)	VS Code Marketplace (https://marketplace.visualstudio.com/items?itemName=toboy.cwtools-vscode)	For editing Paradox game files in Visual Studio Code. It has syntax-highlighting, error checking, auto-completion and more. Based on the CWTools library (https://github.com/cwtools/cwtools) for parsing, editing, and validating Paradox Interactive script files.
Paradox Language Support	Editor (IntelliJ)	JETBRAINS Marketplace (https://plugins.jetbrains.com/plugin/16825-paradox-language-support)	The IntelliJ IDEA plugin for Paradox Interactive game modding. It has syntax-highlighting, error checking, auto-completion and more.
EU5 Language for Notepad++	Editor (Notepad++)	Github (https://github.com/Aetherum17/EU5_Notepad_Language)	Provides EU5-specific script keywords and text highlighting in Notepad++
Arcanum	Map Modding, Visual Editor	GitHub (https://github.com/The-Arcanum-Project/Arcanum)	A feature rich mod and map data editor
Community Mod Framework	Framework, Toolkit, Visual Editor	GitHub (https://github.com/Europa-Universalis-5-Modding-Co-op/community-mod-framework)	A community made mod framework providing a mod menu, custom alerts, action bar, development tools (visual editor, workshop uploader, mod translator), and more.
PDX DeepL	Translation	GitHub (https://github.com/kaiser-chris/pdx-deep-l)	A tool to do incremental auto translation of mods for Paradox games using DeepL
PDX Workshop Manager	General	GitHub (https://github.com/kaiser-chris/pdx-workshop-manager)	A tool to publish Paradox mods, to the Steam workshop in repeatable and configurable way
Pdx-Unlimiter	General	GitHub (https://github.com/crschnick/pdx_unlimiter)	A powerful and smart savegame manager, it can edit save files. It can preview modded flags without starting the game.
UWPDumper	General	GitHub (https://github.com/Wunkolo/UWPDumper)	A tool to extract files from Microsoft Store games.
Clausewitz Maya Exporter	General	Paradox Forums	A tool to create and export 3D models to use in EU5 and other Clausewitz games.
Jomini.js	General	GitHub (https://github.com/nickbabcock/jomini)	A powerful parsing library in javascript based on webassembly, has a C version named Rakaly (https://github.com/rakaly/librakaly).
Europa Universalis V Community Graphical Assets	General	GitHub (https://github.com/Europa-Universalis-5-Modding-Co-op/graphical-assets)	A graphical assets repository for EU5 modders to use and contribute to
EU5 Map Maker	Map modding	Webpage (https://eumapeditor.neurel.ch/)	An online graphical map editor
Location Definition Tool	Map modding	Github (https://github.com/JammingEnd/Eu5_LocationDefinitionTool)	A downloadable graphical map editor



Common problems

■ Some local mod files fail to load.

■ Symptoms:

- A mod is correctly shown in the mod manager, it is added to the playset, but some files do not have an effect.
- `error.log` contains a line such as `[virtualfilesystem.cpp:562]: Could not create lexer from file due to 'fopen() failed: 2'` followed by a line about encoding.
- Potentially also problems with opening `error.log` by clicking on the error deer in game launched with the `-debug_mode` flag.

■ Likely causes:

- Permissions. The game could have trouble accessing the mod files due to broken permissions.
- File path containing non-ASCII characters. It seems like Europa Universalis V (similarly to other games such as Hearts of Iron IV) has troubles with this. On Windows, this can happen due to the user's username, and therefore the user folder containing `Documents/Paradox Interactive/Europa Universalis V/`, containing non-ASCII characters. You can easily check whether this is the case by looking at your file path.

■ Solutions:

- Unfortunately, it looks like modifying the path used by the game for user files (similarly to how it is explained for Hol4) does not seem to work as of writing.
- Symlinking the mod directory to another place also does not work, because the game would still access the original file path.
- For now, if your user folder on Windows contains non-ASCII characters, the simplest solution is to create a new user profile with an ASCII compliant name, and work on your mods from that profile. Renaming the user folder under Users is not advised due to inherent risks of breaking your Windows installation.

Terminology

Europa Universalis 5 modding has a number of terms that are derived from or similar to those in general programming. The following is a partial list of common terms.

Script

This is the 'code' that modders can use to change how the game functions.

Effect

Script commands that change the game state in some way

Trigger

Script checks of the game state that are used to condition effects

Scope

The context for a trigger or effect

Scope link

A keyword for a scope or game value; also called *event target*

The 'class' for a game object



the key for a country

See also

- [Console commands](#)

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"<https://eu5.paradoxwikis.com/index.php?title=Modding&oldid=33186>" sayfasından alınmıştır

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